

ABSTRACT

The world is facing a shortage of energy and it has sent shock waves from Europe to Asia and Pakistan is no exception. Fuel prices have skyrocketed as a result of the crisis between Ukraine and Russia, jeopardizing the supply chain and making it difficult for Pakistan to sustain the efficient functioning of its power plants. Pakistan is undergoing through a severe energy crisis in the recent years. The gap between energy supply and demand has alarmingly widened as the country continues to experience a steep rise in the population and urbanization, ultimately contributing to an amplified demand.

The energy crisis has also had a significant negative impact on the manufacturing sector. It has hampered the production processes of numerous significant and small-scale industries. The industry's supply of gas and electricity was cut off because of the ongoing energy restriction. The South Asian nation is going through a terrible economic crisis, and widespread inflation, a weakening rupee, and declining foreign exchange reserves are making it difficult to purchase energy.

The energy industry in Pakistan is in crisis, due to a lack of energy output to keep up with the country's rising demand during the past few decades. Pakistan is now reliant on imported energy resources like gas and oil. The disparity between Pakistan's total unrestricted gas demand and supply is now four billion cubic feet per day (BCFD), as opposed to the nation's total supply of four BCFD. It goes without saying that throughout the winter, demand quickly rises. In a time when the nation's natural gas reserves were running low, LNG is the most affordable alternative fuel and the only immediate solution.

In order to appease this energy shortage, Pakistan began LNG (Liquefied Natural Gas) imports since 2015. Due to the country's underutilization and insufficient natural reserves exploration, Pakistan is projected to have an increased dependency on LNG imports in the future. As a result of recent conflict between Russia and Ukraine, It is very difficult for developing countries to secure a LNG contract. LNG corporations have violated their agreements with Pakistan in order to exploit the lucrative European markets. Long-term LNG suppliers cancelled a number of cargoes that were due to be delivered during the past few months,

further constricting supplies and creating issues for Pakistan. Currently, Pakistan is being forced to purchase expensive LNG. In view of current LNG market situation, Pakistan needs to organize and develop a comprehensive mechanism which helps in securing a long-term low costs contracts with LNG producer's countries.

Pakistan wants LNG for domestic and industrial use, but do not have enough terminals to process it. This is part of what leads to gas shortages and has in these past few months of winter resulted in massive domestic gas load-shedding. Countries like Qatar are currently interested in investing for LNG terminals establishment and operations.

This report aims to forecast the rise in LNG demand for the upcoming years and the subsequent impact on the handling capacity of Port Qasim Authority (PQA) as the only receiver of these imports. Furthermore, this report will also explore the challenges and infrastructural modifications required to accommodate higher LNG imports and the role of distribution authorities like Sui Southern Gas Company (SSGC).

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1. INTRODUCTION

1.1. BACKGROUND OF THE STUDY

Natural gas has been a primary source of fuel supply in Pakistan, since the country has greater reserves of natural gas over oil sources. However, the production of natural gas in the recent years has become stagnant and the reserves are on a decline. Despite the high demand of natural gas, several factors are contributing to its decline including security concerns over mining and exploration of gas reserves, pricing issues and wastages due to extensive overuse and poor infrastructure (Energy Resource Guide - Pakistan - Oil and Gas, n.d.-a).

Pakistan's power generation relies heavily on the mix of Natural (Non- renewable) sources such as Coal, Oil and Natural Gas and a smaller percentage of renewable sources of energy. Yet, Pakistan relies more heavily on oil and LNG for a majority of its fuel needs, where a significant percentage of oil and LNG is imported due to the country's inability to meet the increasing demands for electricity and power generation. Furthermore, LNG provides greater efficiency in comparisons with other liquid fuels, low maintenance costs, easy transportation, no contamination or adulteration. It is a more viable option to overcome production costs and efficiency of energy generation through oil or coal (Planning & Development, 2019).

1.1.1. PAKISTAN OIL & GAS STATISTICS AND ENERGY SHORTAGES

Pakistan is increasingly dependent on oil imports due to lack of domestic oil refineries to process crude oil. Despite having a capacity to process 19 million tons of Oil, its complete potential has not been explored due to several constraints including financial and technological incompetency. This has severely reduced crude oil production as per 2019 statistics and Pakistan was only able to produce 20% of the total demand which was only around 4.3 million metric tons.(Energy Resource Guide - Pakistan - Oil and Gas, n.d.-b)

Table 1, indicates that Pakistan has only about 2 years of oil left without imports to meet the current consumption levels. It could be observed that Pakistan could only produce around 88,262 Barrels of oil / day which is insufficient to meet the current demand of

country. (“Pakistan Oil Reserves, Production and Consumption Statistics - Worldometer,” 2016)

DESCRIPTION	BARRELS / DAY	GLOBAL RANK
Oil Reserves	353,500,000	52 nd in the world
Oil Production	88,262	53 rd in the world
Oil Consumption	556,000	33 rd in the world
Daily Deficit	-467,738	
Oil Imports	140,097	
Oil Exports	4,896	
Net Imports	135,201	

Table 1 - Pakistan Oil Reserves, Production and Consumption Statistics

Source: (“Pakistan Oil Reserves, Production and Consumption Statistics - Worldometer,” 2016)

Similarly, Natural Gas accounts for 38% of the total energy generation sources. The total gas production and supply is estimated to be around 4 billion cubic feet per day while, the actual demand lies between 6-8 billion cubic feet per day. (Worldometer, 2020).

DESCRIPTION	Million Cubic Ft (MMcft / day)	GLOBAL RANK
Gas Reserves	24,700,000	29 th in the world
Gas Production	1,454,978	26 th in the world
Gas Consumption	1,436,261	21 st in the world
Yearly Surplus	+ 18,717	
Gas Imports	48,382	
Gas Exports	0	
Net Imports	48,382	

Table 2 - Pakistan Gas Reserves, Production and Consumption Statistics

Source: (Worldometer, 2020)

According to the statistics provided in Table 2, the country is only left with 12 years of Natural Gas at its current consumption rates. Also, it could be observed that Pakistan could only produce around 1,454,978 MMCft / day which is insufficient to meet the current demand of country.(Worldometer, 2020).

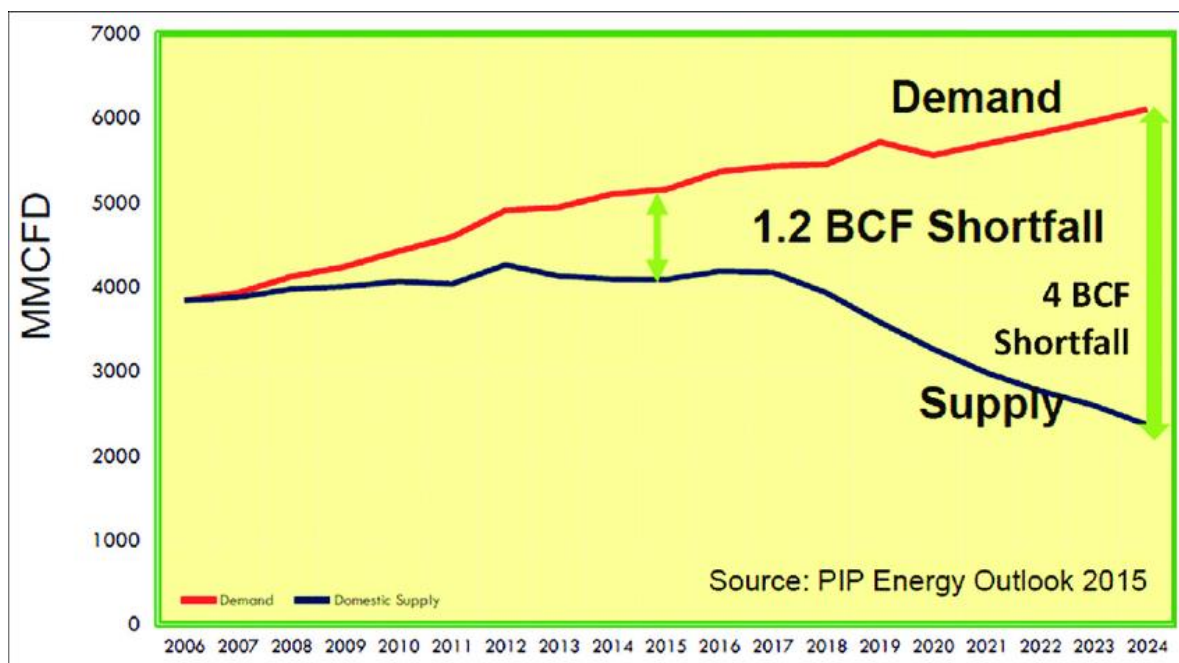


Figure 1-Natural Gas demand and supply curve

Source: (Sajid et al., 2018)

Hence, this has created a severe demand and supply gap to meet the energy requirements of the country due to which Pakistan is facing severe natural gas shortages for the past few years, and as a result it has started relying on Liquefied Natural Gas (LNG) as a source of energy production. LNG imports have significantly risen in Pakistan to cater the demand and supply gap since 2015. Pakistan is relatively a new player in LNG sector and; but, has quickly become a major importer.(Sajid et al., 2018)

1.1.2. LNG STATISTICS

In 2014, the demand and supply gap of the natural gas widen to 3000 million cubic Feet per day. To meet the demand and comply with the severe shortage, the government licensed Pakistan State Oil (PSO) as the authorized buyer of LNG, who would then sell it to Sui Northern Gas Pipeline Limited (SNGPL) for the purpose of regasification, sales and distribution to the customer (Qureshi, 2018a).

While, the Oil and Gas Regulatory Authority (OGRA) is responsible for the up and down stream regulations of the Natural Gas sector. They are the primary authority in issuing licenses for the design, construction, distribution and operation of the LNG infrastructure all over Pakistan.(Qureshi, 2018a).

PSO has imported over 400 LNG cargos, since its authorization in 2015. This amounts to over 27 million metric tons of LNG (Pakistan State Oil Company Limited, n.d.). According to the Pakistan Energy Outlook Report published by the Ministry of planning, development and special incentives Government of Pakistan in March 2022, for the years 2021-2030 shows gas consumptions in various sectors like the fertilizer industry and fuel generation. This will only continue to show an increasing trend. Based on the rising population and urban growth, domestic usage will be the top consuming sector for Natural Gas apart from the industrial sector. Ultimately resulting in more LNG imports to full fill that demand. The consumption trend is forecasted in the figure below.(Planning & Development, 2019).

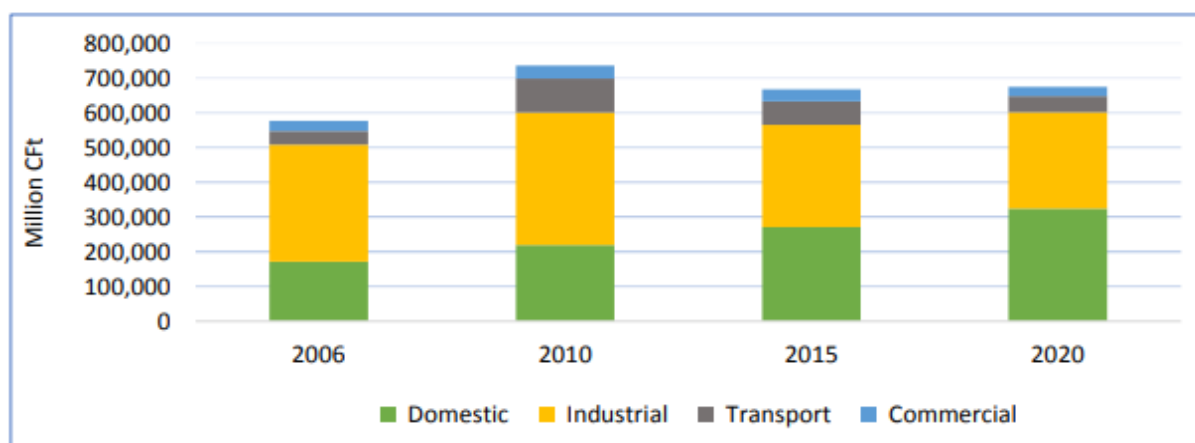


Figure 2 - Gas Consumption by Sector

Source: (Planning & Development, 2019)

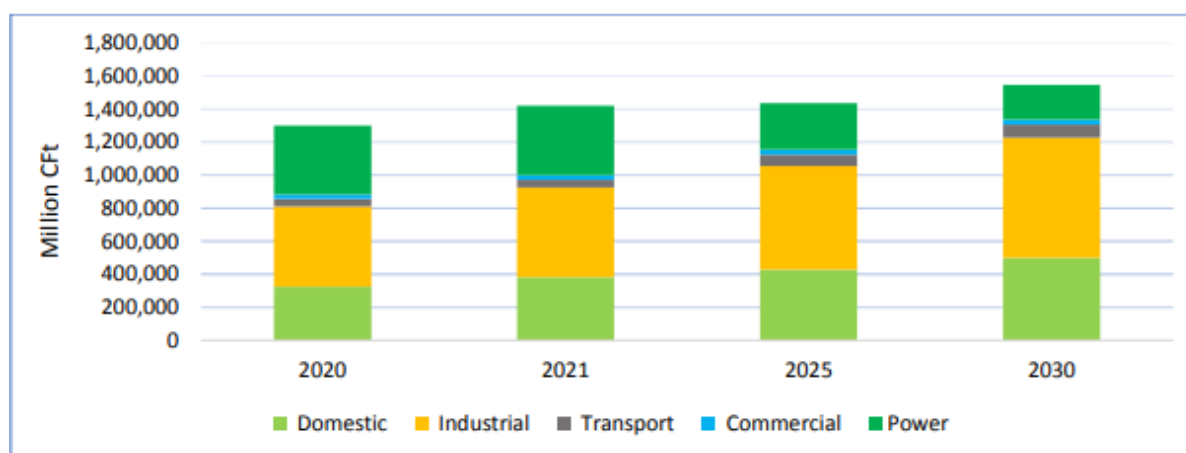


Figure 3 - Energy Forecast for the Natural Gas

Source: (Planning & Development, 2019)

Figures 2 & 3, show a substantially rising forecast in the natural gas consumption for the coming years compared to the gross overall usage before the year 2020. The total gas energy usage for all the sectors combined was only 590,000 MCFT. It further increased to 1,450,000 MCFT in 2021 and is expected to reach an all-time high consumption at 1,590,000 MCFT in 2030.(Planning & Development, 2019)

Furthermore, the Outlook Report also states that Natural Gas and imported LNG now contributes 40% to total energy generation mix. The figure below shows the rise in LNG demand since 2015 until forecasts for 2030. It can be observed that the demand for natural gas supply is insufficient to meet the primary supply. Therefore, the figure below estimates the LNG quantities required to fill the gap and shortages in gas supply (Planning & Development, 2019).

Supply	2006	2010	2015	2020
Natural gas production	1,400,026	1,482,847	1,465,760	1,316,635
Imported LNG	-	-	20,192	355,577
Primary supply	1,400,026	1,482,847	1,485,952	1,672,212

Supply	2020	2021	2025	2030
Net Supply	982,089	942,156	767,392	593,793
Imported LNG	318,241	477,776	669,670	952,680
Primary supply	1,300,330	1,419,931	1,437,062	1,546,473

Table 3 - Gas Supply (Million CFt)

Source: (Planning & Development, 2019)

Pakistan will continue to experience growing shortages in natural gas supply. There will always be varying demand for LNG in the coming years owing to the seasonal variance in demand and consumption; including high spikes during the winter seasons and relatively reduced demands in the summers for both domestic and industrial usages. According to various economic and energy surveys, the LNG imports will skyrocket to 3 times that of the current quantities (Planning & Development, 2019).

Table 3, shows LNG imported for 2020, 2021 and future LNG forecasted imports. Since, there are only two LNG receiving terminals established at PQA with a combined capacity for 1200-1400 MMCFD, it will be insufficient to accommodate the additional imports (Planning & Development, 2019). Therefore, more LNG terminals and supporting infrastructure must be established to manage higher LNG imports.

1.1.3. **LNG SUPPLY CHAIN**

Natural Gas is cooled down at a temperature of -260° Fahrenheit into liquid state. It is easier to store and transport in liquefied form. In its cooled state the volume of the gas shrinks down to about 600 times that in the gaseous state, hence bulk quantities can be easily transported through pipelines. Transportation in its gaseous form is relatively difficult due to greater leakage and specialized channels for transportation. While, it is easier to lay a network of pipelines to transport LNG (Transportation of Liquefied Natural Gas - Energy Education, n.d.).

LNG tankers, also called LNG carriers are super cooled cryogenic tankers that are shipped through the sea by the LNG suppliers and received at PQA LNG terminals. The liquefied gas is then re-gasified in its natural gaseous state, known as RLNG (Re-gasified Liquefied Natural Gas) and then supplied to its consumers by Sui Southern Gas Company (SSGC).

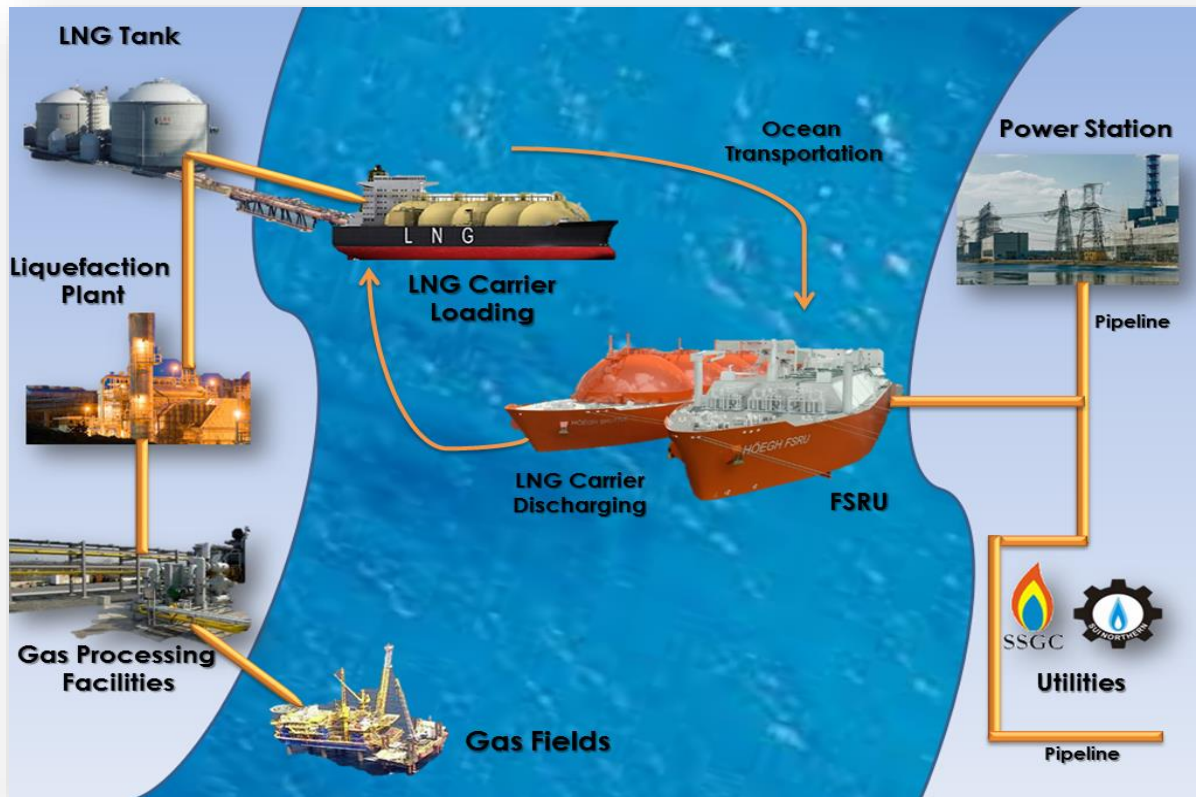


Figure 4 - LNG Supply Chain

Source:(Presentation for PIP 18 Dec 2014, n.d.)

In 2014, SSGC signed an LNG Service Agreement (LSA) with Elengy Terminal Pakistan Limited (ETPL) for the re-gasification, storage and official distribution of LNG throughout the country, with an initial regasification capacity of 200 MMCFD. This was rapidly enhanced to 400-600 MMCFD within 2 years, ultimately producing more than 1600 MW power. There is continuous development of LNG projects at SSGC as a potential replacement for other expensive liquid fuels such as Liquefied Petroleum Gas (LPG).(How SSGC's RLNG Project Will Change Pakistan's Energy Dynamics | Sui Southern Gas Company Limited, n.d.)

1.1.4. **LNG SUPPLY HIERARCHY AND RESPONSIBILITIES**

- PSO, sanctioned by the Government of Pakistan is the sole authorized LNG buyer. It is responsible for an uninterrupted LNG procurement. PSO has 4 procurement possibilities including: **G2G negotiations with Qatar Gas, G2G negotiations with Petronas Malaysia, Spot Purchasing for short term purchasing, EOI with a few years of supply agreement.**
- EETL and PGPCl terminals at Port Qasim, are responsible for receiving LNG imports, store and re-gasify LNG and deliver it to SNGPL.
- SNGPL (Sui Northern Gas Pipeline Limited), is the local RLNG buyer and supplier to SSGC.
- SSGC (Sui Southern Gas Company), receives RLNG from SNGPL and is the official distributor of RLNG to the consumers.

1.1.5. **THE LNG VALUE CHAIN**

The flow chart below describes the LNG supply chain and cash flow distribution at the Engro Elengy Terminal (EETPL) and Pakistan Gas Port Consortium Limited (PGPCL).

Roles	LNG Terminals
LNG Buyer	Pakistan State Oil (PSO) and Pakistan LNG Limited (PLL)
Terminal Operator	Engro Elengy Terminal Pvt. Limited (EETPL) Pakistan Gas Port Consortium Limited (PGPCL)
Terminal Customer	Sui Southern Gas Company Limited (SSGCL)
RLNG Transporter	Sui Southern Gas Company Limited (SSGCL)
RLNG Owner	Sui Northern Gas Pipeline Limited (SNGPL)

Table 4 - LNG Value Chain

Source: (Presentation for PIP 18 Dec 2014, n.d.)

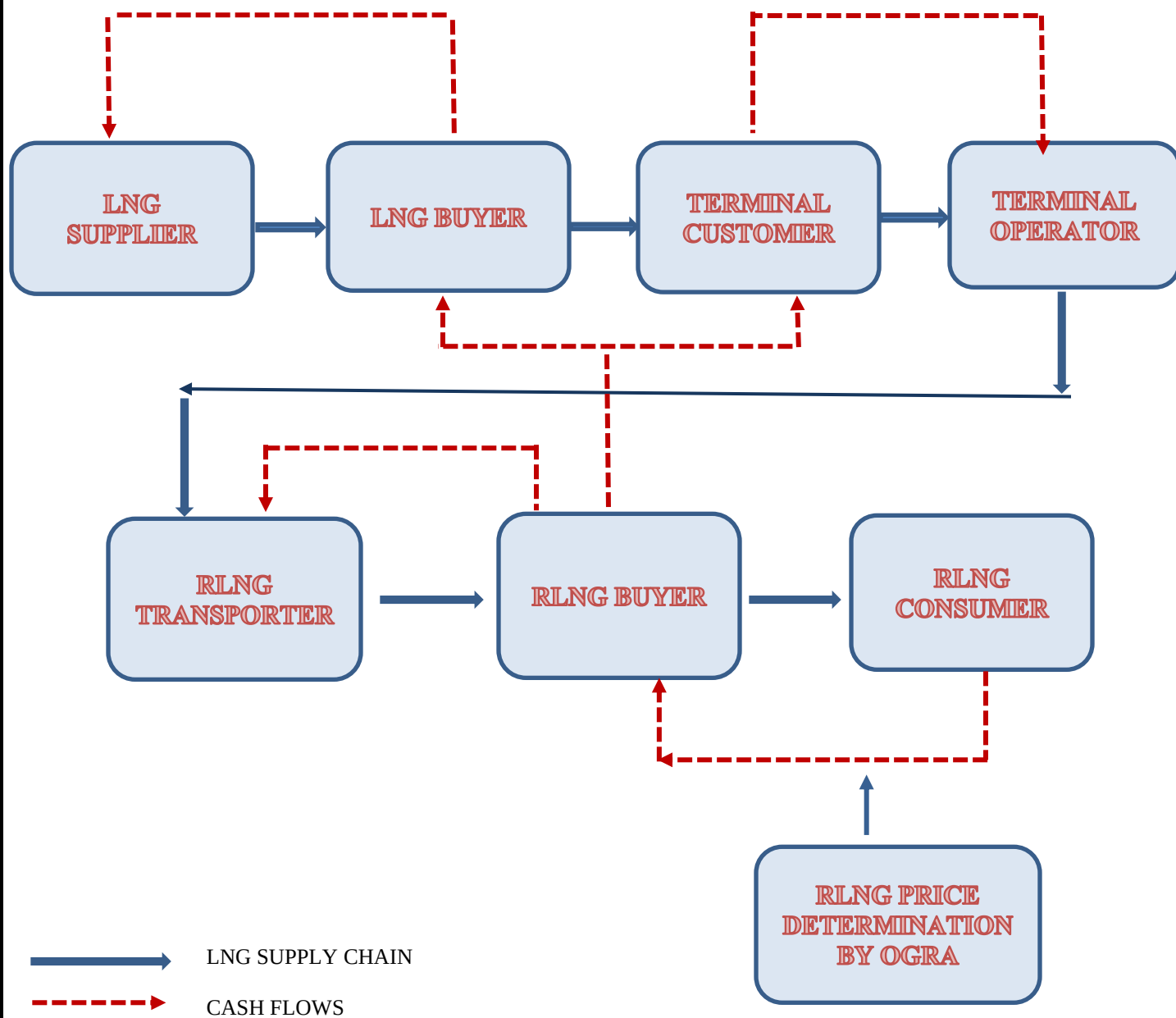


Figure 5 - LNG Value Chain

Source: (Presentation for PIP 18 Dec 2014, n.d.)

1.1.6. LAYOUT OF LNG CARRIERS

LNG carriers are designed using double hulled construction for additional strength to support 4 or 5 larger LNG tanks, usually made of aluminum or 5% nickel that hold LNG. These tanks are heavily insulated with multiple layers to prevent leakage and colder temperatures, needed to keep the gas liquefied. The insulation also prevents the gas from boiling or evaporating during loading and unloading. The LNG tanks are located in the inner hull. A wide space is kept between the inner and the outer hull to prevent heat transferring and damaging the LNG carriers. The tanks in the inner hull are also separated by dry compartments.

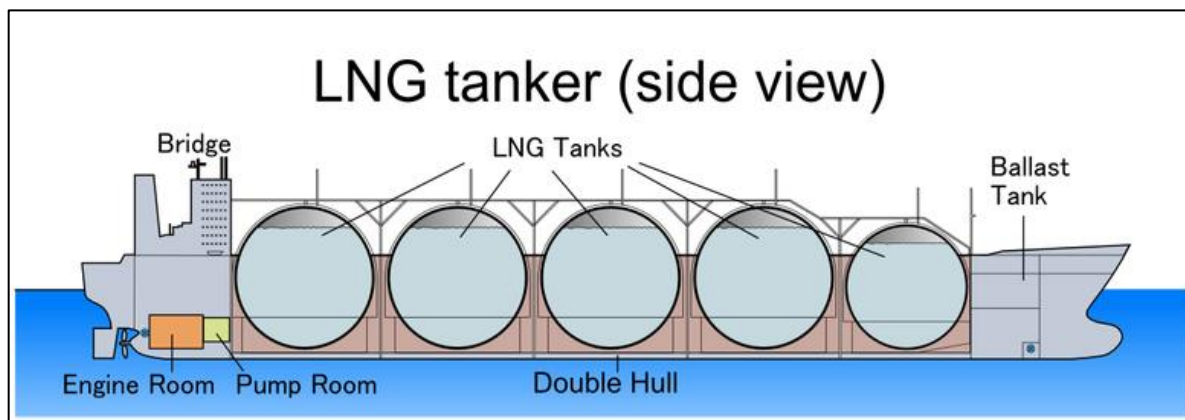


Figure 6 - Layout of an LNG carrier

Source: (Transportation of Liquefied Natural Gas - Energy Education, n.d.)

1.1.7. PORT OASIM AUTHORITY INTRODUCTION

On June 29, 1973, a law passed by the parliament for establishment of Port Qasim. It is situated in the Indus delta region, in the Phitti Creek of Karachi. With an Iron Ore Coal Berth specifically designated for processing bulk imports of raw materials for Pakistan Steel Mill, the port began operations in 1980. Currently, the port has 18 berths, 15 of which are in the commercial sector, with an annual handling capacity of 89 million tons. Modern terminals in the port handle all commodities, including LNG, general cargo, containers, LNG, minerals oils, edible oils, coal, rice, wheat, and cement. It serves as Pakistan's sole LNG port and energy hub.

With regard to all modes of transportation, Port Qasim is well connected to the hinterland. It has immediate access to the national highway, the motorway system that leads to Afghanistan, the CAR Routes, and the CPEC Routes. A 14 km railway line and dedicated railway station also connects the Port to the National Railway network, Port operations are continuous.

Port Qasim has a lot of room for growth. There are currently plans for two new LNG terminals, one container port, and two multifunctional berths. Future plans call for extending and deepening the main navigational channel as well as opening a backup route. It is also planned to dualist the 26 km major access road, upgrade the industrial waste treatment facility that is already in place, and build a new mixed effluent treatment facility.

1.1.8. **LNG TERMINALS ESTABLISHMENT AT POA**

In the wake of this energy crisis, Pakistan established its first LNG receiving terminal in 2015 called the “Engro Elengy floating, storage and regasification terminal” at Port Qasim Authority (PQA). While, its second terminal Pakistan GasPort Consortium Limited terminal (PGPCL) was established 2 years later in 2017 to accommodate further imports as the demand increased. As the demand for LNG further increases, current structure is causing import delays, port congestion, inefficient distribution, and operational issues at the port.

TERMINAL	BERTH	ANNUAL CAPACITY/ MILLION TONS	VESSEL SIZE/ DRAUGHT	COST (US\$)	YEAR OF ESTABLISHMENT
EETPL LNG TERMINAL	1	4.5	259m / 12m	120	2015
PGPCL LNG TERMINAL	1	4.5	259m / 12m	135	2017

Table 5 - LNG Terminals

Source: (Delivering Beyond Expectations _ Engro Elengy Terminal, n.d.-b)

ENGRO ELENGY TERMINAL (PRIVATE) LIMITED

Figure 7 - EETPL LNG Terminal

Source: (Engro Needs a Year for next LNG Import Terminal Decision - Profit by Pakistan Today, n.d.)

The establishment of this port terminal was a joint venture between Engro Corporation (56% share hold) and Royal Vopak of the Netherlands (44% share hold). It is one of the fastest built terminals in the world. The project was completed in only 335 days(Delivering Beyond Expectations _ Engro Elengy Terminal, n.d.-a).

- It has 50% of Pakistan's LNG import and regasification capacity (4.5 million Tons of LNG per year).
- Terminal capacity utilization i.e., regasification ranges between 590 to 630 million cubic feet per day as per customer demand. This caters to about 15% of Pakistan's daily domestic requirements.
- Storage capacity of 150,900 cubic meters.

PAKISTAN GASPORT CONSORTIUM LIMITED TERMINAL (PGPCL)

Figure 8 - PGPCL LNG Terminal

Source: (Pakistan GasPort _ AG, n.d.)

Pakistan GasPort Consortium Limited Terminal established at Mazhar Point; Port Qasim Authority (PQA) is owned by Pakistan GasPort Limited (PGPL) in its entirety's (Pakistan GasPort _ AG, n.d.).

- PGPCL has a 12.3 Km subsea and onshore pipeline that runs between the jetty to the National Gas Grid.
- It has a terminal capacity for regasification of 750 million cubic feet per day.
- The terminal is expected to be a primary fuel source for an estimated 3,600 megawatts for new power generation plants.
- It is further expected to produce about \$2 billion in annual foreign-exchange savings through fuel substitution.

1.2. **RESEARCH GAPS**

Undoubtedly, a lot of research has been conducted on LNG and its forecast specifically and its impact on future prospects and growth but targeting an entity in Pakistan who is the only receiver of LNG at its terminals has not been significantly found.

Considering that, the team has decided to research on LNG Operational Issues at Port Qasim Authority (PQA); Forecast and its Impact on Volume Imported and LNG Carriers Traffic. Studies through statistical models are established and various factors relating to the LNG traffic are highlighted throughout this research.

1.3. **STATEMENT OF THE PROBLEM**

Pakistan is experiencing shortage of gas, creating a mismatch between demand and supply. Hence, to satisfy the increased demand of gas and energy, LNG imports are expected to increase in future.

PQA, being the only receiver port of LNG imports is currently facing problems to accommodate higher volumes of LNG carriers/ vessels at its port, ultimately contributing to port congestion due to lack of infrastructure for LNG handling. Hence, the study is conducted to account such factors and provide solutions for the same.

1.4. **OBJECTIVES OF STUDY**

The purpose of our research is to:

- Study and analyze the existing LNG import statistics and / or infrastructure facilities in Pakistan and forecast future LNG demands through seasonal indexing.
- To analyze the LNG carriers received on two terminals of PQA:
 - a) Engro Elengy Terminal Pakistan Ltd (EETPL)
 - b) Pakistan GasPort Consortium Limited (PGPCL) terminal
- To explore the impact of increased LNG imports and its subsequent traffic at Port Qasim Authority (PQA) in the upcoming years.
- Recommendations and suggestions on improved handling of LNG shipment at PQA will be given by the team members.

1.5. **RESEARCH QUESTIONS**

This research study aims to answer the following research questions:

- What would be the LNG demand in the future?
- How will PQA accommodate the increased demand in its existing facilities and what infrastructural changes would be required?

1.6. **SCOPE OF THE STUDY**

This study aims to identify the vessel traffic, the Port Qasim Authority (PQA) will have to navigate in response to the increased LNG imports in the coming years. The report also utilizes past LNG data that has been obtained through Sui Southern Gas Company (SSGC) for understanding the past market trends to forecast the future LNG quantities.

There are also mathematical inferences through seasonal indexing and regression models which is limited to the port and vessel data available on PQA official website. Even though, LNG distribution is on a national level, our studies and deductions will involve authorities primarily based in Karachi.